

1M1B AI SUSTAINABLE DEVELOPMENT PRESENTATION

in collaboration with IBM SkillsBuild & AICTE



EcoTrack AI

AI-Powered Campus Waste & Sustainability Assistant

TOPIC

EcoTrack AI — Campus Waste & Sustainability Assistant

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Why EcoTrack AI is needed

“How might we use AI to help students and campus communities identify, segregate, and understand waste so that campus waste management can become more sustainable?”



The Gap on Campus

- Students often struggle to identify different types of waste and how to dispose of them correctly.
- Campus waste patterns are hard to understand without simple, accessible digital tools.
- Awareness gaps lead to poor segregation and reduced recycling effectiveness.



Our Aim

EcoTrack AI aims to make waste awareness and sustainability guidance easier and more accessible — turning everyday questions like ‘what bin does this go in?’ into instant, practical answers for students.

Mapped to a Global Goal

SDG

12

**Responsible
Consumption &
Production**



Better Waste Identification

Helping users correctly recognize the type of waste they're handling.



Proper Waste Segregation

Guiding correct sorting at the source, before disposal.



Responsible Disposal Practices

Promoting safe, correct disposal for each waste category.



Waste Reduction Awareness

Building habits that lower overall waste generation.



Sustainability Education

Turning everyday campus moments into learning opportunities.

SOLUTION OVERVIEW

What is EcoTrack AI?

EcoTrack AI is a student-focused sustainability application that uses AI to help campus communities identify waste, understand proper disposal practices, and make more sustainable decisions — built as a Streamlit web app powered by IBM Granite models on watsonx.ai.



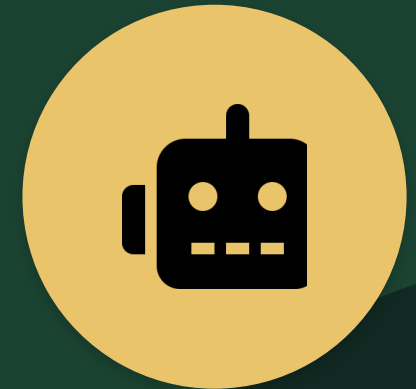
AI Waste Analyzer

Upload a photo — get an instant waste category + disposal guidance.



Sustainability Assistant

Ask sustainability questions, get concise, practical answers.



*Simple. Practical.
Student-friendly AI.*

CORE FEATURE 01

AI Waste Analyzer



How It Works

- 1 Upload a waste image
- 2 IBM Granite Vision analyzes the image when live AI is configured
- 3 The application provides an approximate waste category
- 4 It provides practical disposal guidance
- 5 Confidence / uncertainty is shown when available

Note: When live Vision AI is unavailable, the app uses a clearly labelled Demo Mode — never a fake AI result.

Waste Categories



Plastic



Paper / Cardboard



Food / Organic



E-Waste / Electronic



General Waste

CORE FEATURE 02

AI Sustainability Assistant



The AI Sustainability Assistant lets users ask questions related to sustainability and waste management. When configured, IBM Granite provides concise and practical guidance in plain language.

“

How should plastic bottles be disposed of?

“

How can I reduce waste in my hostel?

“

What is e-waste?

“

How can students reduce food waste?

Note: Live AI responses require the appropriate IBM watsonx.ai configuration.

Beyond the Core Tools



Sustainability Impact Calculator

Provides estimated environmental impact indicators based on user inputs — turning everyday actions into a rough sense of impact.

Results are estimates for educational purposes — not measured or certified environmental results.



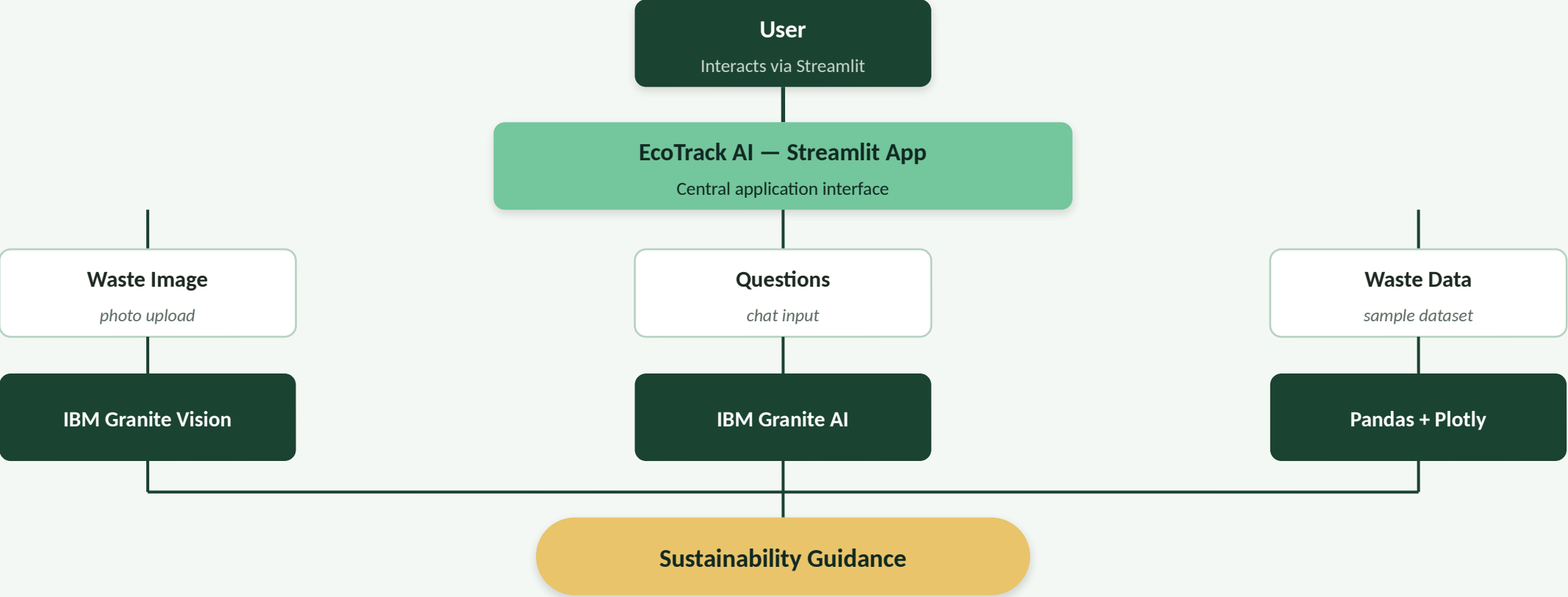
Waste Analytics

Analyzes sample campus waste data using data-processing and visualization tools (Pandas + Plotly) to surface simple patterns and trends.

The included dataset is sample / demo data and does not represent actual campus measurements.

SYSTEM WORKFLOW

How It Works, End to End



TECHNOLOGY STACK

Built With



Python

Application development



Streamlit

Web interface



IBM Granite

AI capabilities



IBM watsonx.ai

AI model access



Pandas

Data processing



Plotly

Data visualization



Pillow

Image processing

MANDATORY SECTION

Responsible AI Considerations



Fairness

Avoiding unnecessary assumptions about users.



Transparency

Clearly identifying AI-generated results.



Ethics

Avoiding fabricated statistics and unsupported claims.



Privacy

Avoiding unnecessary collection of personal information.



Limitations

Recognizing that AI-generated results may be incorrect.

Who Benefits, and How



Target Users

- Students living in hostels & campuses, unsure how to segregate waste.
- Campus sustainability clubs and administration tracking waste habits.
- Faculty and mentors introducing responsible-AI concepts through a live tool.
- Anyone wanting quick, judgement-free sustainability guidance.



Expected Impact

- Improved waste segregation accuracy through instant, on-demand guidance.
- Greater student awareness of sustainable habits, reinforced over time.
- A simple, reusable starting point for real campus-wide waste tracking.
- A model for responsible, transparent use of AI in everyday student life.

Limitations & What's Next



Limitations

- AI-generated responses may contain errors.
- Image classification is approximate.
- Demo Mode is not real AI inference.
- The included dataset is sample data.
- Environmental impact calculations are estimates.
- Official campus waste-management instructions should be followed where applicable.



Future Improvements

- Using real campus waste data.
- Improving waste-image recognition.
- Adding campus-specific disposal guidance.
- Adding long-term waste tracking and reporting.



Thank You

EcoTrack AI — Small tools, real sustainability habits.

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